

ISAAC A. JENSEN

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EDUCATION

Oregon State University

B.S. Computer Science, Systems | GPA: 3.50 (Cum Laude) | 2017–2021

EMPLOYMENT

Software Engineer | Trellix (formerly McAfee) | Sep. 2021 – Present

- Engineer on Trellix IAM platform — identity infrastructure serving **110k+ daily authentications** across all Trellix products, commercial AWS, and FedRAMP GovCloud.
- Owned IAM FedRAMP/GovCloud end-to-end — features, compliance, incident response, FedOps deployment coordination into FedRAMP Moderate, High, DoD IL5.
- Built Keycloak's first edge rate-limiting layer (API Gateway + AWS WAF, CloudFormation IaC), commercial and GovCloud — **de-risked production Keycloak cutover; 1,000 req/min-per-IP WAF rule** plus capacity-based API Gateway throttle.
- Manage **EKS (Kubernetes) clusters** running Keycloak and the entire FedRAMP IAM solution — Helm deployments, rolling upgrades, and Kubernetes Secrets. Sole Keycloak maintainer (3+ yrs) — **10+ zero-downtime upgrades (KC 21 → 26.x)** across custom SPIs, Node.js Lambda clients, and the browser JS SDK.
- Pioneered AI-assisted platform engineering: built Claude Code skill automating Keycloak upgrades from **~2 hours to ~5 minutes** (updates KC + custom SPIs, runs tests, auto-fixes simple issues, escalates larger ones to dev with explanation); AI-driven security review caught real production vulns (SMS brute-force, PII logging, CORS).
- Ported C++ AuthZ service to Node.js/Express via a purpose-built Claude Code skill — **~90% time saved** (E2E in ~2 months vs projected multi-year rewrite); reviewed full flow, fixed niche logic AI missed.
- Remediated **60+ security-scan findings** (Wiz, BlackDuck, Invicti); hardened container images (Iron Bank, RapidFort, Chainguard); upgraded BouncyCastle FIPS libs for FIPS 140-2 & 140-3 compliance.

Software Engineer, Intern | Vacasa | Jun. 2021 – Aug. 2021

- Hardened AWS EC2 instances per Tavora penetration-test findings; fixed front-end UI defects.

Software Engineer, Intern | McAfee | Mar. 2020 – Sep. 2020

- Built Concourse CI pipelines with metadata consolidation into emailed report; wrote net-new JUnit test framework for untested AWS Lambdas — **100% coverage**, enabled team to write own tests.

NOTABLE PROJECTS

- **OAuth Implicit → Authorization Code + PKCE Migration** (Trellix, 2023–2024) — Led platform-wide migration to Authorization Code Flow with PKCE across 4 repos; now default auth flow for **110k+ daily authentications**. **Cut sign-in load ~20%** by removing a blocking, multi-second /scopes fetch that gated the login form — moved scope resolution to a cached backend AuthZ service. *Node.js, OAuth 2.0, Keycloak, JavaScript*
- **SAML IdP Delegation Service** (Trellix, 2026) — Built a reusable SAML IdP service (IAM token → signed SAML assertion + signing-cert metadata API for SP verification) — **unblocked the sale of Trellix's Mobile Threat Defense product**, which couldn't ship without IAM-token→SAML support; any SAML-only product now onboards without custom work. *Node.js, AWS Lambda, Secrets Manager, SAML 2.0*
- **Client Credentials Management Dashboard** (Trellix, 2022–2023) — Full-stack OAuth 2.0 machine-to-machine credential system; built entire backend Lambda API. Replaced ticketed IAM provisioning with on-demand self-service — **cut manual client-creation requests to zero**. *Node.js, AWS Lambda, RDS, AngularJS*
- **Keycloak FIPS 140-2 Prototype** (Trellix, 2021–2022) — Prototyped FIPS 140-2 compliance for Keycloak via BouncyCastle FIPS providers at JVM level; **contributed upstream, cited in official Keycloak release notes**. *Java, BouncyCastle FIPS*

TECHNOLOGIES

- **Languages & Tools:** JavaScript, TypeScript, Java, Python 3, C/C++, Bash, SQL, Node.js, Angular, Express, JUnit, Cypress, Git, Helm
- **Cloud & DevOps:** AWS (Lambda, API Gateway, WAF, CloudFormation, DynamoDB, RDS, EC2, EKS, S3), AWS GovCloud, Docker, Kubernetes, Terraform, Jenkins, CI/CD
- **Identity & Security:** Keycloak, OAuth 2.0 (Auth Code + PKCE / Client Credentials / Implicit), SAML 2.0, OIDC, FIPS 140-2 & 140-3, FedRAMP (Moderate / High), DoD IL5, NIST 800-53